

CH: 9 UNSUPERVISED LEARNING

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Supervised Learning

Supervised learning is a type of machine learning where an algorithm is trained on labelled data, meaning the input data is paired with the correct output. The goal is for the model to learn from this data and make accurate predictions on new, unseen data. **Key Concepts:**

- **Labelled Data:** The training data consists of input-output pairs (e.g., images labelled as "cat" or "dog").
- **Training:** The model learns by mapping inputs to the correct outputs through iterative learning.
- **Loss Function:** Measures how far the model's predictions are from the true outputs. The model minimizes this function during training.
- **Prediction:** After training, the model can predict outputs for new inputs.

Steps in Supervised Learning

1. **Data Collection:** Gather a labelled dataset with input-output pairs (e.g., images with labels, sales data with outcomes).
2. **Data Preprocessing:** Clean and prepare the data by handling missing values, normalizing features, and splitting into training and test sets.
3. **Feature Selection/Extraction:** Identify the most relevant features or create new features that will help the model make accurate predictions.
4. **Model Selection:** Choose an appropriate supervised learning algorithm (e.g., linear regression, decision tree, SVM, etc.) based on the problem type (classification or regression).
5. **Model Training:** Feed the training data into the chosen algorithm, allowing it to learn patterns from the data by adjusting its parameters.
6. **Model Evaluation:** Test the model using a separate test set to evaluate its performance using metrics such as accuracy, precision, recall, or mean squared error (MSE).
7. **Hyperparameter Tuning:** Optimize the model's hyperparameters (e.g., learning rate, number of layers) to improve its performance.
8. **Prediction:** Use the trained model to make predictions on new, unseen data.
9. **Model Deployment:** Implement the model into a system where it can be used to make real-time predictions or decisions.
10. **Model Monitoring and Maintenance:** Continuously monitor the model's performance over time and retrain it if necessary to maintain accuracy.

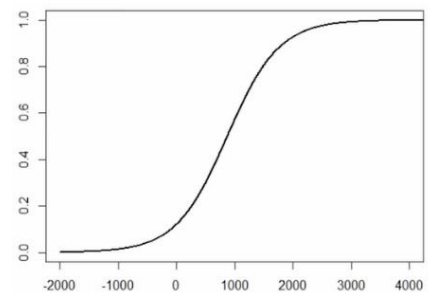
Challenges in Supervised Learning:

1. **Data Labelling:** Requires a large amount of labelled data, which can be expensive and time-consuming to obtain.
2. **Overfitting:** The model may perform well on training data but poorly on new, unseen data if it memorizes rather than generalizes.
3. **Imbalanced Data:** When one class is overrepresented in the data, the model may become biased towards that class.
4. **Scalability:** Training models on very large datasets can be computationally expensive and time-consuming.
5. **Noise in Data:** Errors or inconsistencies in the labelled data can lead to incorrect learning and poor model performance.
6. **Limited Generalization:** A model trained on one dataset may not perform well on a different dataset if the data distributions are different.
7. **Feature Engineering:** Identifying the right features can be difficult and may require domain expertise.

Classification

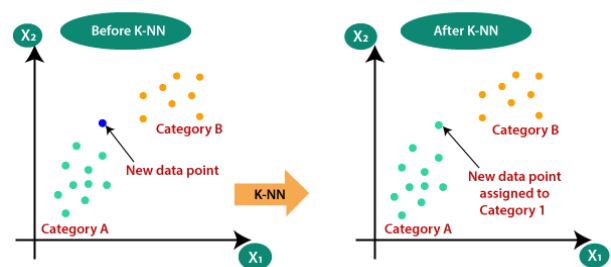
Classification is a type of supervised learning where the goal is to predict a categorical label or class for a given input based on the patterns learned from labelled data. The input is assigned to one of several predefined categories.

1. **Logistic Regression:** Statistical models that use logistic functions to model binary dependent variables. Outputs probabilities and can be used for binary classification tasks. Use information: Email search spam.

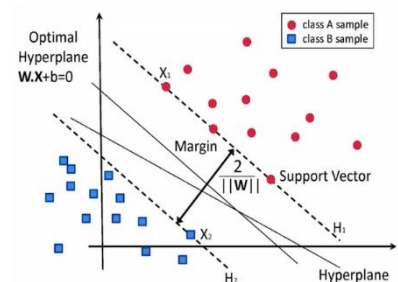


2. **Decision Trees:** Decision tree model. Each node represents a feature (or attribute), each branch represents a rule, and each page represents a result. Reference: Customer segmentation.

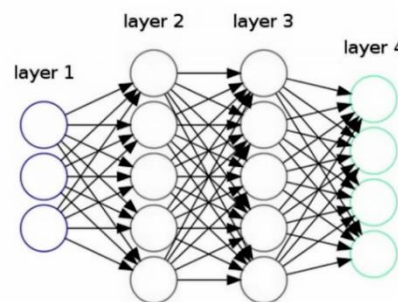
3. **k-Nearest Neighbours (k-NN):** Nonparametric classification technique that assigns labels to points based on their k-nearest training examples within a particular space. Makes use of a distance metric, usually the Euclidean distance, to determine neighbours. Using data: Handwritten digit recognition.



4. **Support Vector Machines (SVM):** The educational supervision model allows it to have the best visibility in separating different classes. Good performance at high altitudes. Data usage: Image classification.



5. **Random Forest:** An ensemble method that combines multiple decision trees to improve accuracy and reduce overfitting. It aggregates the predictions of several trees to make the final decision.
6. **Neural Networks:** Especially with deep learning, neural networks are used for complex classification tasks like image recognition, where input-output relationships are highly non-linear.

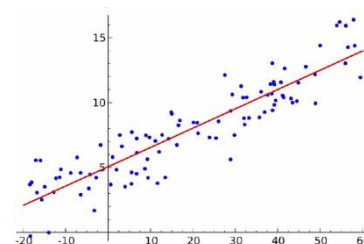


Regression

Regression is a type of supervised learning task that aims to predict a continuous numerical value based on input features. Unlike classification, where the output is discrete, regression deals with estimating relationships among variables and forecasting continuous outcomes.

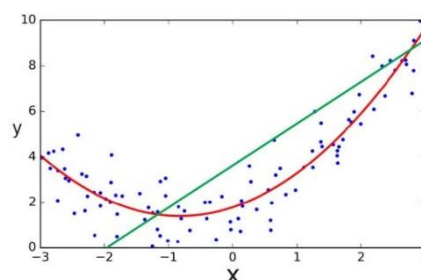
- **Dependent Variable:** The output variable that you are trying to predict (e.g., house prices).
- **Independent Variables:** The input features used to make predictions (e.g., size of the house, location, number of bedrooms).
- **Loss Function:** Measures the difference between the predicted values and the actual values (commonly Mean Squared Error).
- **Training:** The process of adjusting the model parameters to minimize the loss function using historical data.

1. **Linear Regression:** Determine the relationship of dependent with one or more independent variables by fitting a linear equation to observed data. **Use data:** Predict home values based on various characteristics of home values.



2. **Multiple Linear Regression:** An extension of linear regression that uses multiple independent variables to predict a dependent variable. **Use Cases:** Predicting a person's weight based on height, age, and diet.

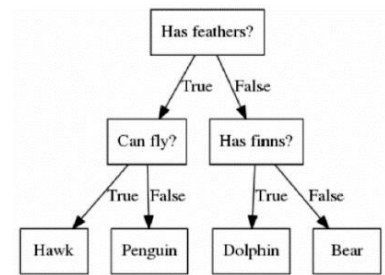
3. **Polynomial Regression:** By specifying polynomial terms of input features, it can achieve linear regression. It allows modelling non-linear relationships between these input features. **Possible use:** model growth rates in biology or finance.



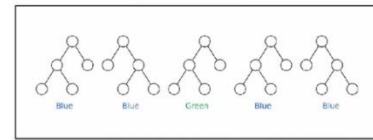
4. **Ridge Regression (L2 Regularization):** A linear regression model that adds a regularization term (penalty) to the cost function to reduce overfitting by minimizing the impact of less significant features
5. **Lasso Regression:** Like Ridge regression, different choices can be made by using the L1 constant and reducing some coefficients to zero. **Use data:** Select the most useful features in the data with multiple estimates.

6. **5. Decision Trees Regression:** The algorithm splits the data into subsets depending on feature value and computes the mean value of the target variable for each subset to construct a model tree.

Usage data: Based on usage patterns and environment to predict device lifetime.



7. **Random Forest Regression:** It is an ensemble approach that involves large numbers of decision trees to multiply the effectiveness of any single prediction. Each tree is trained on a random subset of the data, and the final prediction is the mean prediction of the individual trees.



↓
Blue

"Majority Wins" Model

Unsupervised Learning

Unsupervised learning is a type of machine learning where the model is trained on data without labelled outcomes. The goal is to discover hidden patterns or intrinsic structures within the input data. Unlike supervised learning, there is no predefined output, and the model learns from the raw data itself. **Key Concepts:**

- **No Labels:** Data used in unsupervised learning does not have associated target outputs or labels.
- **Data Exploration:** The focus is on exploring the data and finding relationships, clusters, or patterns.
- **Dimensionality Reduction:** Techniques used to reduce the number of features while retaining essential information.

Applications of Unsupervised Learning

1. **Market Segmentation:** Grouping customers based on purchasing behaviour to tailor marketing strategies.
2. **Anomaly Detection:** Identifying fraudulent transactions or network intrusions by detecting outliers in data.
3. **Recommendation Systems:** Suggesting products or content to users based on similar preferences found in data.
4. **Customer Profiling:** Understanding customer behaviour patterns to improve user experience.
5. **Image Compression:** Reducing the size of images while retaining important features using dimensionality reduction techniques.
6. **Document Clustering:** Grouping similar documents together for information retrieval or topic modelling.

Challenges in Unsupervised Learning

1. **Evaluation:** Lack of ground truth makes it difficult to evaluate the quality of results.
2. **Interpretability:** Understanding and interpreting the patterns or clusters found can be complex.
3. **Scalability:** Some algorithms may struggle to scale effectively with large datasets.
4. **Choosing the Right Algorithm:** Selecting the appropriate technique for a specific task can be challenging without labelled data.
5. **Data Quality:** The effectiveness of unsupervised learning heavily depends on the quality and preprocessing of the input data.

Clustering

Clustering is a type of unsupervised learning that involves grouping a set of objects (data points) into clusters based on their similarities. The objective is to ensure that objects within the same cluster are more like each other than to those in other clusters. Clustering helps in identifying natural groupings in data, making it easier to analyse and interpret. **Key Concepts:**

- **Distance Metrics:** Measures used to quantify the similarity or dissimilarity between data points. Common metrics include Euclidean, Manhattan distance, and cosine similarity.
 - **Centroids:** The central point of a cluster, often used in algorithms like K-Means to represent the cluster.
 - **Inertia:** A measure of how tightly the clusters are packed together(to assess the quality of the clustering).
1. **K-Means Clustering:** Partitions the data into K clusters by assigning each data point to the nearest centroid and updating the centroids based on the mean of the points in each cluster. **Use Cases:** Customer segmentation, image compression. **Limitations:** Requires the number of clusters (K) to be specified in advance and is sensitive to outliers.
 2. **Hierarchical Clustering:** Builds a hierarchy of clusters using either a bottom-up approach (agglomerative) or a top-down approach (divisive). The result is often represented in a dendrogram. **Use Cases:** Taxonomy classification, social network analysis. **Limitations:** Computationally expensive for large datasets.

Aspect	Supervised Learning	Unsupervised Learning
Labeled Data	Requires labeled data (input-output pairs)	Works with unlabeled data (only input)
Goal	Predict or classify outcomes based on past data	Explore the data and find hidden structures or patterns
Output	Produces known outputs like categories or numeric values	No predefined output; identifies inherent structure
Type of Problems	Classification and Regression	Clustering and Association
Training Process	Learns mapping from inputs to outputs	Learns patterns or grouping within input data
Evaluation	Easily evaluated with metrics like accuracy, precision, recall, MSE	Harder to evaluate; often uses visual inspection or silhouette score
Algorithms	Decision Trees, k-NN, SVM, Linear Regression, Neural Networks	K-Means, DBSCAN, Hierarchical Clustering, PCA, Autoencoders
Data Dependency	Performance depends highly on quantity and quality of labeled data	Can work with unlabeled data, but quality still matters
Complexity	Generally more computational due to labeled data processing	Often less complex initially but may require deeper analysis
Examples	Email spam detection, credit scoring, disease prediction	Market segmentation, document clustering, image compression
Accuracy	Can be very accurate with good training data	Accuracy is relative; effectiveness depends on the clarity of discovered structure
Feature Engineering	Features are often engineered based on known outputs	Feature discovery can be automatic or exploratory

Usage Stage	Often used in deployment or production systems (e.g., ML models in apps)	Often used in research, exploration, or pre-processing
Learning Feedback	Learns with feedback (error correction)	No direct feedback during training
Dependence on Output	Strongly depends on the existence of correct output labels	Focuses only on input patterns

Reinforcement Learning

Reinforcement Learning (RL) is a type of machine learning where an agent learns to make decisions by interacting with an environment. The agent takes actions in this environment to maximize a cumulative reward over time. Unlike supervised learning, where the model is trained on labeled data, in reinforcement learning, the agent learns from the consequences of its actions.

- **Agent:** The learner or decision-maker that interacts with the environment.
- **Environment:** The setting in which the agent operates. The agent observes the state of the environment and takes actions.
- **State:** A representation of the current situation of the agent within the environment.
- **Action:** The choices available to the agent to interact with the environment.
- **Reward:** A feedback signal received by the agent after taking an action in a particular state, indicating the success of that action.
- **Policy:** A strategy that the agent employs to determine which actions to take in different states. It can be deterministic or stochastic.
- **Value Function:** A function that estimates the expected cumulative reward that can be obtained from a given state or state-action pair.

Applications of Reinforcement Learning:

1. **Game Playing:** RL has been successfully applied in games like chess, Go, and video games (e.g., AlphaGo and OpenAI's Dota 2 agent).
2. **Robotics:** Training robots to perform tasks such as walking, grasping objects, or navigating.
3. **Autonomous Vehicles:** Making real-time decisions based on sensor inputs to navigate and respond to dynamic environments.
4. **Healthcare:** Personalizing treatment plans for patients based on their responses to different interventions.
5. **Recommendation Systems:** Adapting recommendations based on user interactions and feedback.

Challenges in Reinforcement Learning:

1. **Exploration vs. Exploitation:** Balancing the need to explore new actions (to discover their rewards) with the need to exploit known actions that yield high rewards.
2. **Sample Efficiency:** Many RL algorithms require a large number of interactions with the environment to learn effectively, which can be impractical in real-world applications.
3. **Sparse Rewards:** In some environments, the agent may receive very few rewards, making it difficult to learn effective policies.
4. **Generalization:** Ensuring that learned policies can generalize well to unseen states or environments.
5. **Stability and Convergence:** Ensuring that the learning process is stable and converges to optimal policies can be challenging, especially in complex environments.